

Name of Student: Shreyas Kulkarni	PRN: 25070122225
Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 4	
TITLE : Wrapper Classes in Java	

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Programme

1. Create a program to convert student marks from String format to Integer and calculate total marks.

```
StudentMarks.java
1 import java.util.Scanner;
2
3 class StudentMarks {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         System.out.print("Enter Mark 1: ");
8         String m1Str = scanner.nextLine();
9         System.out.print("Enter Mark 2: ");
10        String m2Str = scanner.nextLine();
11
12        Integer mark1 = Integer.valueOf(m1Str);
13        Integer mark2 = Integer.valueOf(m2Str);
14
15        int total = mark1 + mark2;
16
17        System.out.println("Total Marks: " + total);
18        scanner.close();
19    }
20 }
```

2. Develop an Employee Payroll System that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

```
EmployeePayroll.java
1- import java.util.Scanner;
2
3- class EmployeePayroll {
4-     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         System.out.print("Enter Employee ID: ");
8         Integer empId = Integer.valueOf(scanner.nextLine());
9
10        System.out.print("Enter Basic Salary: ");
11        Double basicSalary = Double.valueOf(scanner.nextLine());
12
13        System.out.print("Enter Bonus Amount: ");
14        Double bonus = Double.valueOf(scanner.nextLine());
15
16        if (basicSalary <= 0 || bonus < 0) {
17            System.out.println("Invalid input: Basic salary must be greater than zero and bonus
                cannot be negative.");
18        } else {
19            double netSalary = basicSalary + bonus;
20            System.out.println("Payroll Details\n");
21            System.out.println("Employee ID: " + empId);
22            System.out.println("Net Salary: Rs." + netSalary);
23        }
24
25        scanner.close();
26    }
27 }
```

Output

Program 1 output:

```
Output
Enter Mark 1: 33
Enter Mark 2: 65
Total Marks: 98

=== Code Execution Successful ===
```

Program 2 output:

```
Output
Enter Employee ID: 144
Enter Basic Salary: 56999
Enter Bonus Amount: 12000
Payroll Details

Employee ID: 144
Net Salary: Rs.68999.0

=== Code Execution Successful ===
```

Conclusion

These programs highlight the use of Java Wrapper Classes (Integer, Double) to parse textual user input into numeric objects. The Student Marks Converter demonstrates converting string inputs into primitive numbers via wrapper methods to perform mathematical operations. The Employee Payroll System builds on this by wrapping user inputs, performing validation checks to prevent negative or zero salary figures, and cleanly computing final payroll values.